

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. in CSE Final Examination, 2022
CSE-2102: Object Oriented Programming

Total Marks: 70

Time: 3 Hours

Answer any 5 (Five) of the following Questions:

1. a) What is meant by OOP? Explain the fundamental principles of OOP and their significance. 3
- b) Define byte code. Give and explain the basic structure of java program. 3
- c) Define class with example. Why is main method static in java? Explain. 3
- d) Consider the following java program written by a novice programmer: 5

```
Class JavaTest extends ClassA, ClassB {
    public void main(String *args){
        System.In.println("This is a simple java program");
    }
}
```

Point out his mistakes. Also correct the program so that it can be compiled.

2. a) Why are the keywords "this" and "static" used in Java? Explain. 4
- b) Write a Java program to find the prime numbers in a given range. 5
- c) Is there any method to call a sub-class constructor from a superclass constructor? 2
- d) Why constructors in Java cannot be static? 3
3. a) What is inheritance? Discuss different types of inheritance with block diagram and example. 4
- b) What is the ambiguity that arises in multiple inheritance? How it can be overcome? Explain with example. 4
- c) Define method overriding with suitable example? What are the rules for method overriding? 3
- d) Define method overloading. Can we overload main() method in java? Explain. 3
4. a) What is Garbage Collection in Java? How does Garbage Collection work in Java? 4
- b) Describe Inner Class & Outer Class with suitable example. 5
- c) What does it mean by abstract class? Why is it needed? Explain. 5
5. a) Discuss dynamic method dispatch with example. 4
- b) Dynamic binding is also referred to as a run-time polymorphism. Expand & Explain. 4
- c) Write a Java program that has a class named shape. In this program there are three sub classes named circle, triangle and square. Here, each class has two member methods named draw() & erase(). 6
6. a) Define generics. What are bounded types in Java generics, and how do they restrict the types that can be used as type arguments? 3
- b) Define inner class. Write a program that will illustrate the concept of method local inner class. 5
- c) Why do you use super keyword? What are the conditions for a class to become abstract? Can you declare an object of abstract class? 3
- d) What is final keyword? Why "this" keyword is used in java language? Explain with example. 3
7. a) Discuss the basic structure of exception handling in java. 3
- b) Explain the use of finally block in java exception handling mechanism. 5
- c) List five common examples of exceptions. 4
- d) Explain the role of the "throw" and "throws" keywords in Java exception handling with proper example. 2